UNIVERSITAS NEGERI YOGYAKARTA
FAKULTAS TEKNIK
PROGRAM STUDI TEKNIK INDUSTRI

LAPORAN PRAKTIKUM

APLIKASI WEB DAN MOBILE

Semester Genap 2025/2026

MODUL PERTEMUAN 9 – Pengenalan React.js (Component, JSX, Props)

Nama Mahasiswa	:	Syalfa Oktifa Nanda
NIM	:	23051430023
Kelas	:	A
Tanggal Praktikum	:	24 April 2026
Dosen Pengampu	:	Dr. Eng. Ir. Aji Ery Burhandenny, S.T., M.AIT.

Dikumpulkan paling lambat 7 hari setelah pelaksanaan praktikum

A. IDENTITAS & INFORMASI MODUL

Mata Kuliah	Aplikasi Web dan Mobile
Kode Modul	Modul Pengenalan React.js (Component, JSX, Props) Pertemuan 9
Topik Utama	Pengenalan React.js (Component, JSX, Props)
Sub-Topik	Setup Lingkungan React, Konsep Component, JSX Syntax, dan Props (Data Passing).
Durasi Praktikum	340 Menit
Tanggal Praktikum	24 April 2026
Nama Mahasiswa	Syalfa Oktifa Nanda
NIM	23051430023
Kelas	A

B. TUJUAN PEMBELAJARAN

1	Memahami konsep Library vs Framework dan mengapa React dipilih untuk aplikasi skala besar.
2	Mampu menyiapkan environment pengembangan React menggunakan Vite.
3	Memahami sintaks JSX (JavaScript XML).
4	Mampu membuat Functional Component yang menerima dan menampilkan data melalui Props.

C. ALAT DAN BAHAN

No.	Nama Alat / Bahan / Aplikasi	Versi / Spesifikasi	Keterangan
1	Visual Studio Code (VS Code)	Versi 1.8x ke atas	Code Editor
2	Google Chrome / Browser Modern	Versi terbaru	Testing & Debugging
3	Node.js & npm	Node 18+ / npm 9+	Runtime JS (Modul tertentu)
4	React.js	Versi terbaru	Library frontend
5	Vite	Versi 5.x	Build tool React

D. DASAR TEORI

D.1 Konsep Utama

React adalah library JavaScript yang digunakan untuk membangun tampilan antarmuka pengguna (UI) secara dinamis dan efisien. React menggunakan konsep component-based, yaitu membagi tampilan menjadi bagian-bagian kecil yang dapat digunakan kembali. Hal ini membuat pengembangan aplikasi menjadi lebih terstruktur dan mudah dikelola.

Salah satu fitur utama React adalah JSX (JavaScript XML), yang memungkinkan penulisan HTML di dalam JavaScript. Dengan JSX, developer dapat membuat tampilan UI dengan lebih mudah karena sintaksnya mirip dengan HTML, namun tetap memiliki kekuatan JavaScript.

Selain itu, terdapat konsep props yang digunakan untuk mengirim data dari satu komponen ke komponen lain. Props ini berfungsi seperti parameter pada fungsi, sehingga setiap komponen dapat menerima data yang berbeda dan menghasilkan tampilan yang dinamis.

D.2 Keterkaitan dengan Teknik Industri

Dalam bidang Teknik Industri, React dapat digunakan untuk membuat dashboard monitoring produksi, sistem manajemen data karyawan, dan visualisasi data operasional. Misalnya, dalam sebuah pabrik, setiap mesin dapat direpresentasikan sebagai komponen React yang menampilkan status mesin, jumlah produksi, dan kondisi operasional secara real-time.

Dengan pendekatan component-based, sistem menjadi lebih mudah dikembangkan dan dirawat, terutama jika jumlah data sangat besar. Hal ini sangat relevan dalam industri modern yang membutuhkan sistem digital untuk meningkatkan efisiensi dan pengambilan keputusan.

E. LANGKAH KERJA DAN HASIL PRAKTIKUM

E.1 Langkah 1 – Setup Project dengan Vite

Deskripsi singkat apa yang dilakukan pada langkah ini:

Pada langkah ini dilakukan pembuatan project React menggunakan Vite melalui terminal di Visual Studio Code. Perintah yang digunakan adalah npm create vite, kemudian dilanjutkan dengan npm install dan npm run dev untuk menjalankan aplikasi.

Namun, pada proses ini terjadi error karena versi Node.js yang digunakan belum memenuhi syarat minimum yang dibutuhkan oleh Vite. Versi Node.js yang digunakan adalah 20.12.2, sedangkan Vite membutuhkan versi minimal 20.19 atau lebih tinggi.

Solusi yang dilakukan:

- Mengganti versi Vite menjadi versi 5 agar kompatibel
- Menyesuaikan plugin React
- Melakukan install ulang dependency

Setelah dilakukan perbaikan, aplikasi berhasil dijalankan dan muncul pada browser dengan alamat http://localhost:5173.

```
/* ======= KODE / OUTPUT LANGKAH 1 ======= */
Node.js v20.12.2
npm notice
npm notice New major version of npm available! 10.5.0 -> 11.13.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.13.0
npm notice Run npm install -g npm@11.13.0 to update!
npm notice
npm ERR! code 1
npm ERR! path C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026
npm ERR! command failed
npm ERR! command C:\Windows\system32\cmd.exe /d /s /c create-vite sistem-industri --template
react

npm ERR! A complete log of this run can be found in: C:\Users\Advan\AppData\Local\npm-
cache\_logs\2026-04-24T01_19_01_965Z-debug-0.log
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026> cd sistem-industri
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
install
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-array@0.23.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-helpers@0.5.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/core@1.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/js@10.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/object-schema@3.0.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/plugin-kit@0.7.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@vitejs/plugin-react@6.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint@10.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
```

```
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
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npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
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npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'rolldown@1.0.0-rc.17',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'vite@8.0.10',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

up to date, audited 136 packages in 2s

31 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite

You are using Node.js 20.12.2. Vite requires Node.js version 20.19+ or 22.12+. Please upgrade your
Node.js version.
file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507
                if (loadErrors.length > 0) throw new Error("Cannot find native binding. npm has a bug
related to optional dependencies (https://github.com/npm/cli/issues/4828). Please try `npm i` again
after removing both package-lock.json and node_modules directory.", { cause: loadErrors.reduce((err,
cur) => {
                                          ^

Error: Cannot find native binding. npm has a bug related to optional dependencies
(https://github.com/npm/cli/issues/4828). Please try `npm i` again after removing both package-
lock.json and node_modules directory.
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507:36
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
    ... 2 lines matching cause stack trace ...
```

```
    at async ModuleLoader.import (node:internal/modules/esm/loader:323:24)
    at async CAC.<anonymous>
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/vite/dist/node/cli.js:707:27) {
  [cause]: Error: Cannot find module '@rolldown/binding-win32-x64-msvc'
  Require stack:
  - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
      at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
      ... 2 lines matching cause stack trace ...
      at require (node:internal/modules/helpers:179:18)
      at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:147:21)
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/parse-B_ZnWxLZ.mjs:3:46
      at ModuleJob.run (node:internal/modules/esm/module_job:222:25)
      at async ModuleLoader.import (node:internal/modules/esm/loader:323:24) {
    code: 'MODULE_NOT_FOUND',
    requireStack: [
      'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
    ],
    cause: Error: Cannot find module './rolldown-binding.win32-x64-msvc.node'
    Require stack:
    - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
        at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
        at Module._load (node:internal/modules/cjs/loader:984:27)
        at Module.require (node:internal/modules/cjs/loader:1231:19)
        at require (node:internal/modules/helpers:179:18)
        at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:142:12)
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
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      code: 'MODULE_NOT_FOUND',
      requireStack: [
        'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
      ]
    }
  }
}

Node.js v20.12.2
```

```
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
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npm WARN EBADENGINE Unsupported engine {
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npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
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npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
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npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-visitor-keys@5.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
```

```
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'rolldown@1.0.0-rc.17',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'vite@8.0.10',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

added 135 packages, and audited 136 packages in 3s

31 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite

You are using Node.js 20.12.2. Vite requires Node.js version 20.19+ or 22.12+. Please upgrade your
Node.js version.
file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507
                if (loadErrors.length > 0) throw new Error("Cannot find native binding. npm has a bug
related to optional dependencies (https://github.com/npm/cli/issues/4828). Please try `npm i` again
after removing both package-lock.json and node_modules directory.", { cause: loadErrors.reduce((err,
cur) => {
                                         ^

Error: Cannot find native binding. npm has a bug related to optional dependencies
(https://github.com/npm/cli/issues/4828). Please try `npm i` again after removing both package-
lock.json and node_modules directory.
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507:36
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
    ... 2 lines matching cause stack trace ...
    at async ModuleLoader.import (node:internal/modules/esm/loader:323:24)
    at async CAC.<anonymous>
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/vite/dist/node/cli.js:707:27) {
  [cause]: Error: Cannot find module '@rolldown/binding-win32-x64-msvc'
  Require stack:
  - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
      at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
      ... 2 lines matching cause stack trace ...
      at require (node:internal/modules/helpers:179:18)
      at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:147:21)
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
```

```
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/parse-B_ZnWxLZ.mjs:3:46
      at ModuleJob.run (node:internal/modules/esm/module_job:222:25)
      at async ModuleLoader.import (node:internal/modules/esm/loader:323:24) {
    code: 'MODULE_NOT_FOUND',
    requireStack: [
      'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
    ],
    cause: Error: Cannot find module './rolldown-binding.win32-x64-msvc.node'
    Require stack:
    - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
        at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
        at Module._load (node:internal/modules/cjs/loader:984:27)
        at Module.require (node:internal/modules/cjs/loader:1231:19)
        at require (node:internal/modules/helpers:179:18)
        at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:142:12)
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/parse-B_ZnWxLZ.mjs:3:46
        at ModuleJob.run (node:internal/modules/esm/module_job:222:25)
        at async ModuleLoader.import (node:internal/modules/esm/loader:323:24) {
      code: 'MODULE_NOT_FOUND',
      requireStack: [
        'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
      ]
    }
  }
}

Node.js v20.12.2
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
install vite@5 --save-dev
npm WARN ERESOLVE overriding peer dependency
npm WARN While resolving: sistem-industri@0.0.0
npm WARN Found: vite@8.0.10
npm WARN node_modules/vite
npm WARN   peer vite@"^8.0.0" from @vitejs/plugin-react@6.0.1
npm WARN   node_modules/@vitejs/plugin-react
npm WARN     dev @vitejs/plugin-react@"^6.0.1" from the root project
npm WARN   1 more (the root project)
npm WARN
npm WARN Could not resolve dependency:
npm WARN peer vite@"^8.0.0" from @vitejs/plugin-react@6.0.1
npm WARN node_modules/@vitejs/plugin-react
npm WARN   dev @vitejs/plugin-react@"^6.0.1" from the root project
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-array@0.23.5',
```

```
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-helpers@0.5.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/core@1.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/js@10.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/object-schema@3.0.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/plugin-kit@0.7.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@vitejs/plugin-react@6.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint@10.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-visitor-keys@5.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

added 5 packages, removed 6 packages, changed 1 package, and audited 135 packages in 14s

28 packages are looking for funding
```

```
  run `npm fund` for details

2 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite

failed to load config from C:\Users\Advan\GITHUBERP
SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri\vite.config.js
error when starting dev server:
Error [ERR_PACKAGE_PATH_NOT_EXPORTED]: Package subpath './internal' is not defined by
"exports" in C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\vite\package.json imported from C:\Users\Advan\GITHUBERP
SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri\node_modules\@vitejs\plugin-
react\dist\index.js
    at exportsNotFound (node:internal/modules/esm/resolve:304:10)
    at packageExportsResolve (node:internal/modules/esm/resolve:651:9)
    at packageResolve (node:internal/modules/esm/resolve:837:14)
    at moduleResolve (node:internal/modules/esm/resolve:927:18)
    at defaultResolve (node:internal/modules/esm/resolve:1157:11)
    at ModuleLoader.defaultResolve (node:internal/modules/esm/loader:390:12)
    at ModuleLoader.resolve (node:internal/modules/esm/loader:359:25)
    at ModuleLoader.getModuleJob (node:internal/modules/esm/loader:234:38)
    at ModuleWrap.<anonymous> (node:internal/modules/esm/module_job:87:39)
    at link (node:internal/modules/esm/module_job:86:36)
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
install @vitejs/plugin-react@4 --save-dev
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-array@0.23.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-helpers@0.5.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/core@1.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/js@10.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/object-schema@3.0.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
```

```
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/plugin-kit@0.7.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint@10.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-visitor-keys@5.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

added 8 packages, changed 2 packages, and audited 143 packages in 6s

28 packages are looking for funding
  run `npm fund` for details

2 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite


  VITE v5.4.21  ready in 423 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help Node.js v20.12.2
npm notice
npm notice New major version of npm available! 10.5.0 -> 11.13.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.13.0
npm notice Run npm install -g npm@11.13.0 to update!
npm notice
```

```
npm ERR! code 1
npm ERR! path C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026
npm ERR! command failed
npm ERR! command C:\Windows\system32\cmd.exe /d /s /c create-vite sistem-industri --template
react

npm ERR! A complete log of this run can be found in: C:\Users\Advan\AppData\Local\npm-
cache\_logs\2026-04-24T01_19_01_965Z-debug-0.log
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026> cd sistem-industri
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
install
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-array@0.23.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-helpers@0.5.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/core@1.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/js@10.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/object-schema@3.0.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/plugin-kit@0.7.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@vitejs/plugin-react@6.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint@10.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
```

```
npm WARN EBADENGINE   package: 'eslint-visitor-keys@5.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'rolldown@1.0.0-rc.17',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'vite@8.0.10',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

up to date, audited 136 packages in 2s

31 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite

You are using Node.js 20.12.2. Vite requires Node.js version 20.19+ or 22.12+. Please upgrade your
Node.js version.
file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507
                if (loadErrors.length > 0) throw new Error("Cannot find native binding. npm has a bug
related to optional dependencies (https://github.com/npm/cli/issues/4828). Please try `npm i` again
after removing both package-lock.json and node_modules directory.", { cause: loadErrors.reduce((err,
cur) => {
                                        ^

Error: Cannot find native binding. npm has a bug related to optional dependencies
(https://github.com/npm/cli/issues/4828). Please try `npm i` again after removing both package-
lock.json and node_modules directory.
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507:36
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
    ... 2 lines matching cause stack trace ...
    at async ModuleLoader.import (node:internal/modules/esm/loader:323:24)
    at async CAC.<anonymous>
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/vite/dist/node/cli.js:707:27) {
  [cause]: Error: Cannot find module '@rolldown/binding-win32-x64-msvc'
  Require stack:
```

```
  - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
      at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
      ... 2 lines matching cause stack trace ...
      at require (node:internal/modules/helpers:179:18)
      at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:147:21)
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/parse-B_ZnWxLZ.mjs:3:46
      at ModuleJob.run (node:internal/modules/esm/module_job:222:25)
      at async ModuleLoader.import (node:internal/modules/esm/loader:323:24) {
    code: 'MODULE_NOT_FOUND',
    requireStack: [
      'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
    ],
    cause: Error: Cannot find module './rolldown-binding.win32-x64-msvc.node'
    Require stack:
    - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
        at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
        at Module._load (node:internal/modules/cjs/loader:984:27)
        at Module.require (node:internal/modules/cjs/loader:1231:19)
        at require (node:internal/modules/helpers:179:18)
        at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:142:12)
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/parse-B_ZnWxLZ.mjs:3:46
        at ModuleJob.run (node:internal/modules/esm/module_job:222:25)
        at async ModuleLoader.import (node:internal/modules/esm/loader:323:24) {
      code: 'MODULE_NOT_FOUND',
      requireStack: [
        'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
      ]
    }
  }
}

Node.js v20.12.2
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
install
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-array@0.23.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
```

```
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-helpers@0.5.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/core@1.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/js@10.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/object-schema@3.0.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/plugin-kit@0.7.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@vitejs/plugin-react@6.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint@10.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-visitor-keys@5.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'rolldown@1.0.0-rc.17',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'vite@8.0.10',
```

```
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

added 135 packages, and audited 136 packages in 3s

31 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite

You are using Node.js 20.12.2. Vite requires Node.js version 20.19+ or 22.12+. Please upgrade your
Node.js version.
file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507
                if (loadErrors.length > 0) throw new Error("Cannot find native binding. npm has a bug
related to optional dependencies (https://github.com/npm/cli/issues/4828). Please try `npm i` again
after removing both package-lock.json and node_modules directory.", { cause: loadErrors.reduce((err,
cur) => {
                                          ^

Error: Cannot find native binding. npm has a bug related to optional dependencies
(https://github.com/npm/cli/issues/4828). Please try `npm i` again after removing both package-
lock.json and node_modules directory.
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:507:36
    at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
    ... 2 lines matching cause stack trace ...
    at async ModuleLoader.import (node:internal/modules/esm/loader:323:24)
    at async CAC.<anonymous>
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/vite/dist/node/cli.js:707:27) {
  [cause]: Error: Cannot find module '@rolldown/binding-win32-x64-msvc'
  Require stack:
  - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
      at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
      ... 2 lines matching cause stack trace ...
      at require (node:internal/modules/helpers:179:18)
      at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:147:21)
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
      at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/parse-B_ZnWxLZ.mjs:3:46
      at ModuleJob.run (node:internal/modules/esm/module_job:222:25)
      at async ModuleLoader.import (node:internal/modules/esm/loader:323:24) {
    code: 'MODULE_NOT_FOUND',
```

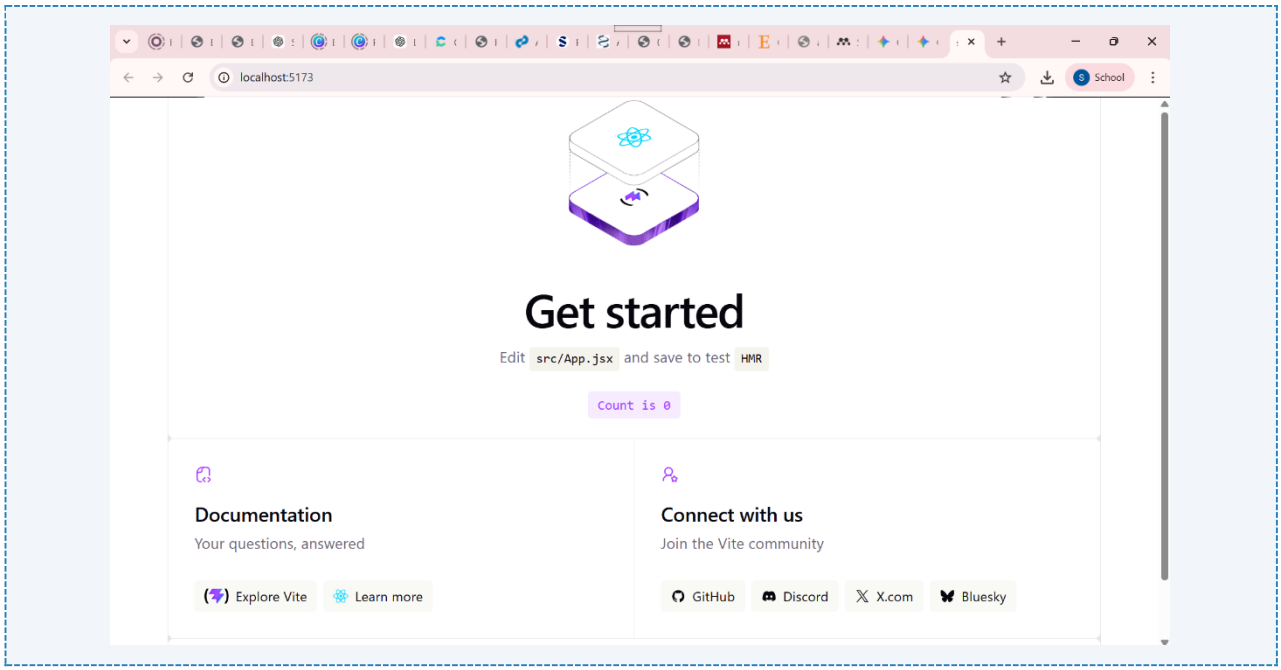
```
    requireStack: [
      'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
    ],
    cause: Error: Cannot find module './rolldown-binding.win32-x64-msvc.node'
    Require stack:
    - C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\rolldown\dist\shared\binding-BeU_1iEk.mjs
        at Module._resolveFilename (node:internal/modules/cjs/loader:1143:15)
        at Module._load (node:internal/modules/cjs/loader:984:27)
        at Module.require (node:internal/modules/cjs/loader:1231:19)
        at require (node:internal/modules/helpers:179:18)
        at requireNative
(file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:142:12)
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:475:18
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/binding-BeU_1iEk.mjs:9:49
        at file:///C:/Users/Advan/GITHUBERP%20SYALFA/PraktikumWebMob_TekInd_2026/sistem-
industri/node_modules/rolldown/dist/shared/parse-B_ZnWxLZ.mjs:3:46
        at ModuleJob.run (node:internal/modules/esm/module_job:222:25)
        at async ModuleLoader.import (node:internal/modules/esm/loader:323:24) {
      code: 'MODULE_NOT_FOUND',
      requireStack: [
        'C:\\Users\\Advan\\GITHUBERP SYALFA\\PraktikumWebMob_TekInd_2026\\sistem-
industri\\node_modules\\rolldown\\dist\\shared\\binding-BeU_1iEk.mjs'
      ]
    }
  }
}

Node.js v20.12.2
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
install vite@5 --save-dev
npm WARN ERESOLVE overriding peer dependency
npm WARN While resolving: sistem-industri@0.0.0
npm WARN Found: vite@8.0.10
npm WARN node_modules/vite
npm WARN   peer vite@"^8.0.0" from @vitejs/plugin-react@6.0.1
npm WARN   node_modules/@vitejs/plugin-react
npm WARN     dev @vitejs/plugin-react@"^6.0.1" from the root project
npm WARN   1 more (the root project)
npm WARN
npm WARN Could not resolve dependency:
npm WARN peer vite@"^8.0.0" from @vitejs/plugin-react@6.0.1
npm WARN node_modules/@vitejs/plugin-react
npm WARN   dev @vitejs/plugin-react@"^6.0.1" from the root project
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-array@0.23.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-helpers@0.5.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
```

```
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/core@1.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/js@10.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/object-schema@3.0.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/plugin-kit@0.7.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@vitejs/plugin-react@6.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || >=22.12.0' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint@10.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-visitor-keys@5.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

added 5 packages, removed 6 packages, changed 1 package, and audited 135 packages in 14s

28 packages are looking for funding
  run `npm fund` for details

2 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force
```

```
Run `npm audit` for details.
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite

failed to load config from C:\Users\Advan\GITHUBERP
SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri\vite.config.js
error when starting dev server:
Error [ERR_PACKAGE_PATH_NOT_EXPORTED]: Package subpath './internal' is not defined by
"exports" in C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-
industri\node_modules\vite\package.json imported from C:\Users\Advan\GITHUBERP
SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri\node_modules\@vitejs\plugin-
react\dist\index.js
    at exportsNotFound (node:internal/modules/esm/resolve:304:10)
    at packageExportsResolve (node:internal/modules/esm/resolve:651:9)
    at packageResolve (node:internal/modules/esm/resolve:837:14)
    at moduleResolve (node:internal/modules/esm/resolve:927:18)
    at defaultResolve (node:internal/modules/esm/resolve:1157:11)
    at ModuleLoader.defaultResolve (node:internal/modules/esm/loader:390:12)
    at ModuleLoader.resolve (node:internal/modules/esm/loader:359:25)
    at ModuleLoader.getModuleJob (node:internal/modules/esm/loader:234:38)
    at ModuleWrap.<anonymous> (node:internal/modules/esm/module_job:87:39)
    at link (node:internal/modules/esm/module_job:86:36)
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
install @vitejs/plugin-react@4 --save-dev
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-array@0.23.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/config-helpers@0.5.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/core@1.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/js@10.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/object-schema@3.0.5',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: '@eslint/plugin-kit@0.7.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
```

```
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint@10.2.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-scope@9.1.2',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'eslint-visitor-keys@5.0.1',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }
npm WARN EBADENGINE Unsupported engine {
npm WARN EBADENGINE   package: 'espree@11.2.0',
npm WARN EBADENGINE   required: { node: '^20.19.0 || ^22.13.0 || >=24' },
npm WARN EBADENGINE   current: { node: 'v20.12.2', npm: '10.5.0' }
npm WARN EBADENGINE }

added 8 packages, changed 2 packages, and audited 143 packages in 6s

28 packages are looking for funding
  run `npm fund` for details

2 moderate severity vulnerabilities

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
PS C:\Users\Advan\GITHUBERP SYALFA\PraktikumWebMob_TekInd_2026\sistem-industri> npm
run dev

> sistem-industri@0.0.0 dev
> vite


  VITE v5.4.21  ready in 423 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

Screenshot Hasil:



Gambar E.1 – Menampilkan proses pembuatan project React menggunakan Vite melalui terminal pada Visual Studio Code.

E.2 Langkah 2 – Membersihkan Project

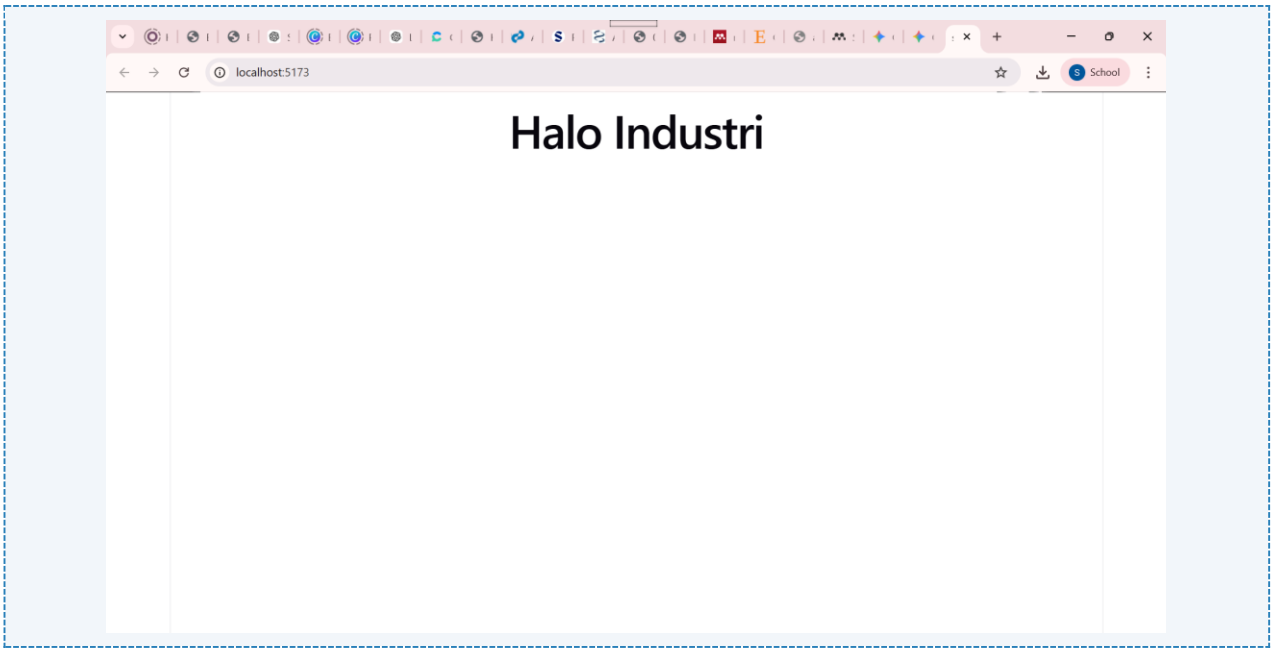
Pada tahap ini dilakukan penghapusan kode default pada file App.jsx untuk menyederhanakan tampilan. Seluruh isi return dihapus dan diganti dengan tampilan sederhana:

<h1>Halo Industri</h1>

Hal ini bertujuan agar fokus pada pembelajaran dasar React tanpa gangguan kode bawaan.

```
/* ======= KODE / OUTPUT LANGKAH 2 ======= */
import React from 'react';

function App() {
  return (
    <h1>Halo Industri</h1>
  );
}

export default App;
```

Screenshot Hasil:



Gambar E.2 – Menunjukkan proses instalasi dependency menggunakan perintah npm install pada terminal.

E.3 Langkah 3 – Membuat Komponen Pertama: KartuMesin

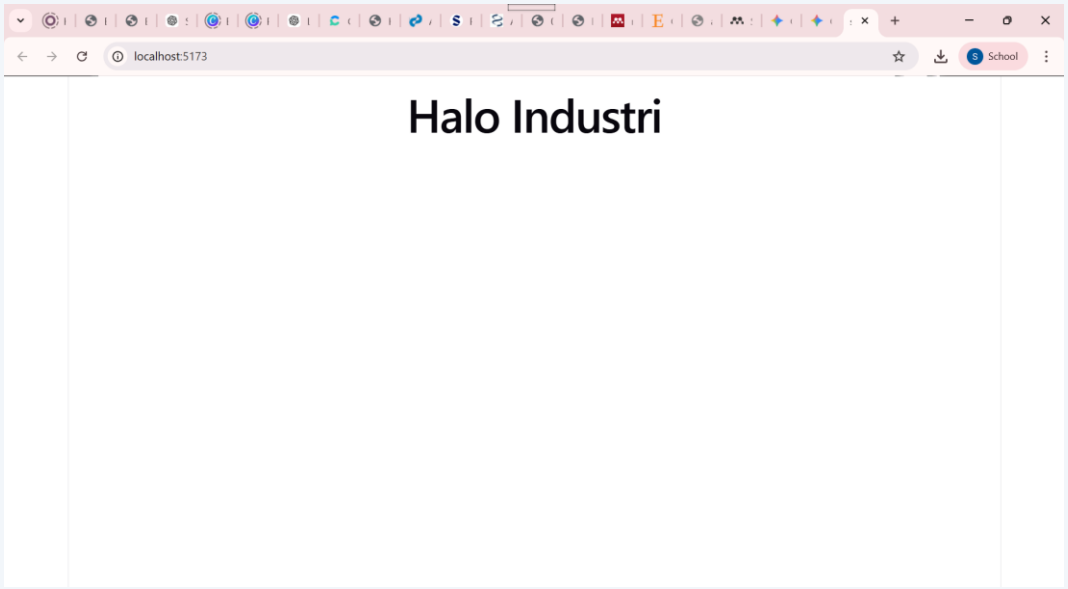
Pada langkah ini dibuat komponen baru bernama KartuMesin.jsx yang berfungsi untuk menampilkan informasi mesin seperti nama, status, dan jumlah produksi.

Komponen ini menggunakan props untuk menerima data, sehingga dapat digunakan berulang kali dengan data yang berbeda.

```
/* ======= KODE / OUTPUT LANGKAH 3 ======= */
// 1. Import React
import React from 'react';
// 2. Membuat Komponen Fungsi yang menerima 'props'
function KartuMesin(props) {
    // Menerima data dari props
    const namaMesin = props.nama;
    const status = props.status;
    const produksi = props.produksi;
    // Logika penentuan warna badge berdasarkan status
    let badgeColor = 'bg-secondary';
    if (status === 'Running') badgeColor = 'bg-success';
    if (status === 'Stop') badgeColor = 'bg-danger';
    if (status === 'Maintenance') badgeColor = 'bg-warning';
    return (
        // Menggunakan class -> className di JSX
        <div className="card shadow-sm p-3 mb-3">
            <div className="card-body">
                <h5 className="card-title">{namaMesin}</h5>
                <span className={`badge ${badgeColor}`}>{status}</span>
                <hr />
```

```
                <p>Produksi Saat Ini: <strong>{produksi}</strong> Unit</p>
            </div>
        </div>
    );
}
// 3. Export agar bisa dipakai di file lain
export default KartuMesin;
```

Screenshot Hasil:



Gambar E.3 – Menampilkan proses menjalankan aplikasi React menggunakan perintah npm run dev.

E.4 Langkah 4 – Menggunakan Komponen di App.jsx

Komponen KartuMesin kemudian dipanggil di dalam App.jsx sebanyak tiga kali dengan data yang berbeda. Setiap komponen menampilkan informasi mesin yang berbeda, seperti:

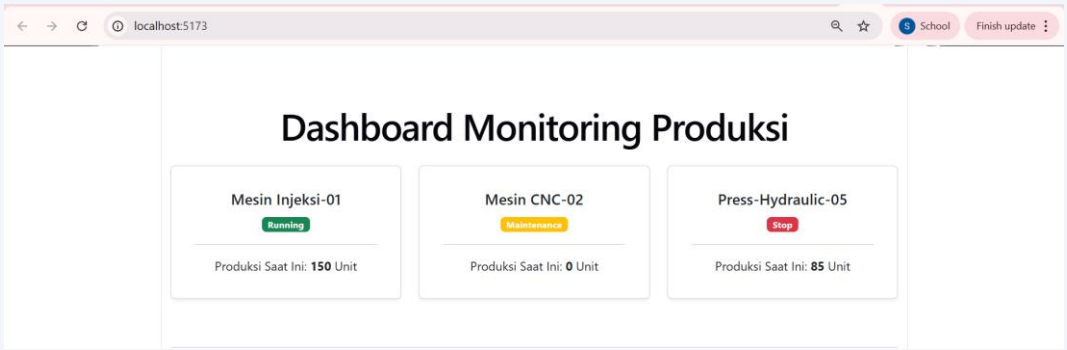
- CNC Turning (Running)
- CNC Milling (Maintenance)
- Press Hydraulic (Stop)

Dengan cara ini, satu komponen dapat digunakan berkali-kali tanpa perlu menulis ulang kode.

```
/* ======= KODE / OUTPUT LANGKAH 3 ======= */
import React from 'react';
// Memanggil komponen yang baru kita buat
import KartuMesin from './Komponen/KartuMesin';

function App() {
```

```
  return (
    <div className="container mt-5">
      <h1 className="text-center mb-4">Dashboard Monitoring Produksi</h1>

      <div className="row">
        {/* Memanggil Kartu Mesin 1 */}
        <div className="col-md-4">
          <KartuMesin
            nama="Mesin Injeksi-01"
            status="Running"
            produksi={150}
          />
        </div>

        {/* Memanggil Kartu Mesin 2 */}
        <div className="col-md-4">
          <KartuMesin
            nama="Mesin CNC-02"
            status="Maintenance"
            produksi={0}
          />
        </div>

        {/* Memanggil Kartu Mesin 3 */}
        <div className="col-md-4">
          <KartuMesin
            nama="Press-Hydraulic-05"
            status="Stop"
            produksi={85}
          />
        </div>
      </div>
    </div>
  );
}

export default App;
```

Screenshot Hasil:



Gambar E.4 – Menunjukkan tampilan awal aplikasi React di browser setelah berhasil dijalankan.

F. LATIHAN DAN TUGAS MANDIRI

F.1 Latihan 1 – Destructuring Props

Pertanyaan / Instruksi Latihan:

> Latihan 1 (Destructuring Props): Ubah cara deklarasi props di `KartuMesin.jsx` menggunakan destructuring: `function KartuMesin({ nama, status, produksi }) { ... }`. Sesuaikan kode di dalamnya.

Jawaban / Kode Solusi:

KartuMesin.jsx

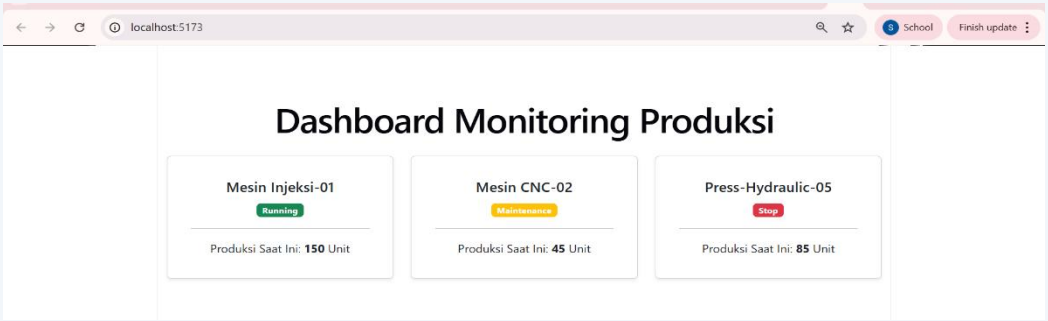
```
import React from 'react';

// Latihan 1: Menggunakan kurung kurawal {} untuk mengambil data langsung
function KartuMesin({ nama, status, produksi }) {

    let badgeColor = 'bg-secondary';
    if (status === 'Running') badgeColor = 'bg-success';
    if (status === 'Stop') badgeColor = 'bg-danger';
    if (status === 'Maintenance') badgeColor = 'bg-warning';

    return (
        <div className="card shadow-sm p-3 mb-3">
            <div className="card-body text-center">
                <h5 className="card-title">{nama}</h5>
                <span className={`badge ${badgeColor}`}>{status}</span>
```

```
                <hr />
                {/* Sekarang langsung panggil {produksi}, bukan
props.produksi lagi */}
                <p>Produksi Saat Ini: <strong>{produksi}</strong> Unit</p>
            </div>
        </div>
    );
}

export default KartuMesin;
```

App.jsx

```
import React from 'react';
import KartuMesin from './Komponen/KartuMesin';
import KartuKaryawan from './Komponen/KartuKaryawan'; // Baris ini sangat
penting!

function App() {
  return (
    <div className="container mt-5">
      {/* BAGIAN MONITORING MESIN (Langkah 4 + Latihan 1 & 2) */}
      <h1 className="text-center mb-4">Dashboard Monitoring Produksi</h1>
      <div className="row mb-5">
        <div className="col-md-4">
          <KartuMesin nama="Mesin Injeksi-01" status="Running" produksi={150}
/>
        </div>
        <div className="col-md-4">
          <KartuMesin nama="Mesin CNC-02" status="Maintenance" produksi={45}
/>
        </div>
        <div className="col-md-4">
          <KartuMesin nama="Press-Hydraulic-05" status="Stop" produksi={85}
/>
        </div>
      </div>

      <hr />

      {/* TUGAS PROYEK MINI: BAGIAN KARYAWAN */}
      <h2 className="text-center my-4">Daftar Karyawan Aktif</h2>
      <div className="row">
        <div className="col-md-4">
          <KartuKaryawan nama="Budi Santoso" jabatan="Manager"
bagian="Produksi" />
        </div>
```

```
        <div className="col-md-4">
          <KartuKaryawan nama="Siti Aminah" jabatan="Operator"
bagian="Assembly" />
        </div>
        <div className="col-md-4">
          <KartuKaryawan nama="Agus Pratama" jabatan="QC" bagian="Quality
Control" />
        </div>
      </div>
    </div>
  );
}

export default App;
```



Gambar F.1 – Menampilkan proses pembuatan project React menggunakan Vite melalui terminal pada Visual Studio Code.

F.2 Latihan 2 – Default Props

Pertanyaan / Instruksi Latihan:

> Latihan 2 (Default Props): Berikan nilai default pada prop `produksi` menjadi 0, jika user lupa mengirim prop tersebut.

Jawaban / Kode Solusi:

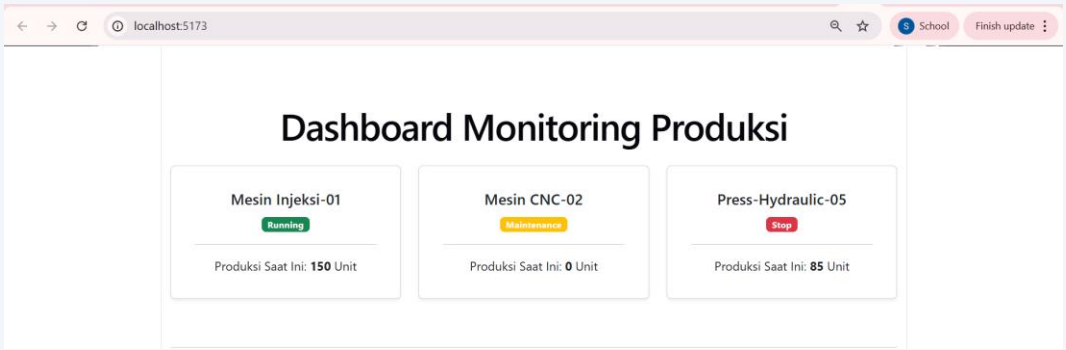
KartuMesin.jsx

```
import React from 'react';

// Di sini kita pakai { nama, status, produksi = 0 } sesuai perintah Latihan
1 & 2
function KartuMesin({ nama, status, produksi = 0 }) {

    let badgeColor = 'bg-secondary';
    if (status === 'Running') badgeColor = 'bg-success';
    if (status === 'Stop') badgeColor = 'bg-danger';
    if (status === 'Maintenance') badgeColor = 'bg-warning';

    return (
        <div className="card shadow-sm p-3 mb-3">
            <div className="card-body text-center">
                <h5 className="card-title">{nama}</h5>
                <span className={`badge ${badgeColor}`}>{status}</span>
                <hr />
                <p>Produksi Saat Ini: <strong>{produksi}</strong> Unit</p>
            </div>
        </div>
    );
}

export default KartuMesin;
```

App.jsx

```
import React from 'react';
import KartuMesin from './Komponen/KartuMesin';
import KartuKaryawan from './Komponen/KartuKaryawan'; // Baris ini sangat
penting!

function App() {
  return (
    <div className="container mt-5">
      {/* BAGIAN MONITORING MESIN (Langkah 4 + Latihan 1 & 2) */}
      <h1 className="text-center mb-4">Dashboard Monitoring Produksi</h1>
      <div className="row mb-5">
        <div className="col-md-4">
          <KartuMesin nama="Mesin Injeksi-01" status="Running" produksi={150}
/>
        </div>
        <div className="col-md-4">
          {/* Latihan 2: Mengosongkan produksi untuk ngetes default value 0
*/}
          <KartuMesin nama="Mesin CNC-02" status="Maintenance" />
        </div>
```

```
        <div className="col-md-4">
          <KartuMesin nama="Press-Hydraulic-05" status="Stop" produksi={85}
/>
        </div>
      </div>

      <hr />

      {/* TUGAS PROYEK MINI: BAGIAN KARYAWAN */}
      <h2 className="text-center my-4">Daftar Karyawan Aktif</h2>
      <div className="row">
        <div className="col-md-4">
          <KartuKaryawan nama="Budi Santoso" jabatan="Manager"
bagian="Produksi" />
        </div>
        <div className="col-md-4">
          <KartuKaryawan nama="Siti Aminah" jabatan="Operator"
bagian="Assembly" />
        </div>
        <div className="col-md-4">
          <KartuKaryawan nama="Agus Pratama" jabatan="QC" bagian="Quality
Control" />
        </div>
      </div>
    </div>
  );
}

export default App;
```



Gambar F.2 – Menunjukkan hasil tampilan awal setelah project berhasil dijalankan di browser.

F.3 Tugas Proyek Mini – Kartu Karyawan

Deskripsi proyek mini:

> Tugas Proyek Mini (Kartu Karyawan):
> □ Buat komponen baru bernama `KartuKaryawan.jsx`. Komponen ini menerima props: `nama`, `jabatan`, `bagian`.
> □ Tampilkan komponen ini 3 kali di `App.jsx` dengan data karyawan berbeda (Manager, Operator, QC).

Penjelasan pendekatan/solusi yang digunakan:

> Dalam menyelesaikan proyek mini ini, pendekatan yang digunakan adalah membuat komponen terpisah bernama KartuKaryawan.jsx yang berfungsi untuk menampilkan data karyawan. Komponen ini dirancang menggunakan konsep functional component dan menerima data melalui props yaitu nama, jabatan, dan bagian.
>
> Setelah komponen dibuat, langkah selanjutnya adalah mengimpor komponen tersebut ke dalam file App.jsx. Di dalam App.jsx, komponen KartuKaryawan dipanggil sebanyak tiga kali dengan data yang berbeda, yaitu untuk posisi Manager, Operator, dan Quality Control (QC).
>
> Pendekatan ini dipilih karena sesuai dengan konsep React yang berbasis komponen, sehingga kode menjadi lebih rapi, mudah dibaca, dan dapat digunakan kembali tanpa harus menulis ulang struktur tampilan.

Kode Utama:

KartuKaryawan.jsx

```
import React from 'react';

// Latihan 3: Membuat komponen KartuKaryawan
function KartuKaryawan({ nama, jabatan, bagian }) {
    return (
```

```
        <div className="card border-primary p-3 mb-3 shadow-sm">
            <div className="card-body">
                <h5 className="card-title text-primary">{nama}</h5>
                <p className="mb-1"><strong>Jabatan:</strong> {jabatan}</p>
                <p className="mb-0"><strong>Bagian:</strong> {bagian}</p>
            </div>
        </div>
    );
}

export default KartuKaryawan;
```

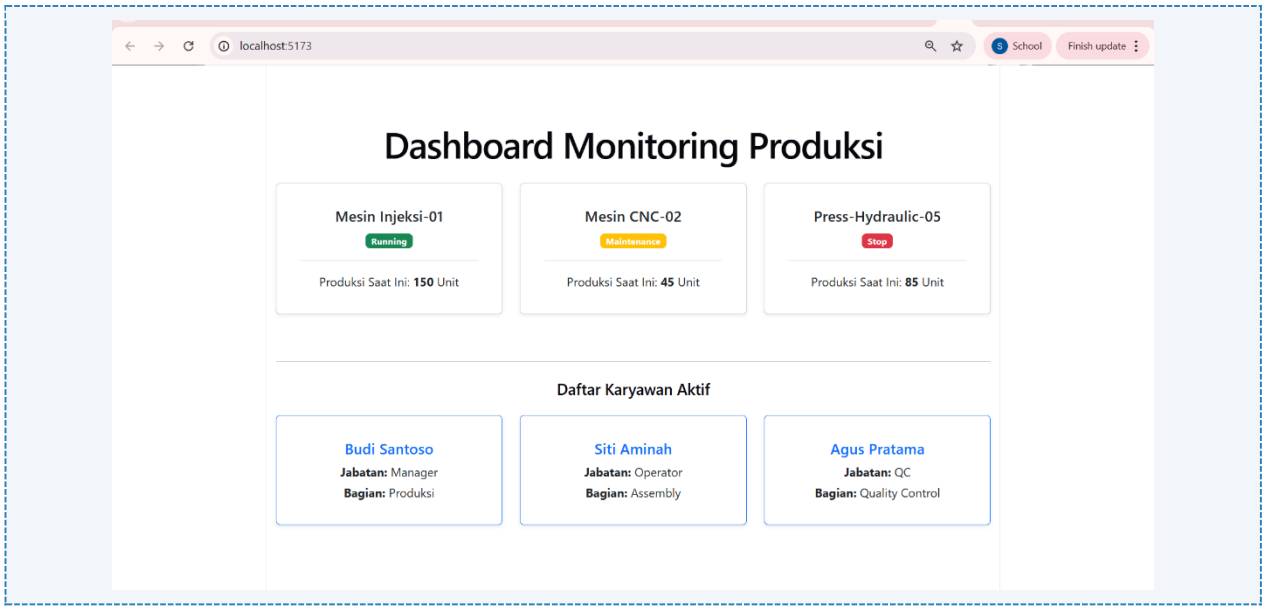
KartuMesin.jsx

```
import React from 'react';

// Latihan 1: Menggunakan kurung kurawal {} untuk mengambil data langsung
function KartuMesin({ nama, status, produksi }) {

    let badgeColor = 'bg-secondary';
    if (status === 'Running') badgeColor = 'bg-success';
    if (status === 'Stop') badgeColor = 'bg-danger';
    if (status === 'Maintenance') badgeColor = 'bg-warning';

    return (
        <div className="card shadow-sm p-3 mb-3">
            <div className="card-body text-center">
                <h5 className="card-title">{nama}</h5>
                <span className={`badge ${badgeColor}`}>{status}</span>
                <hr />
                {/* Sekarang langsung panggil {produksi}, bukan
props.produksi lagi */}
                <p>Produksi Saat Ini: <strong>{produksi}</strong> Unit</p>
            </div>
        </div>
    );
}

export default KartuMesin;
```

App.jsx

```
import React from 'react';
import KartuMesin from './Komponen/KartuMesin';
import KartuKaryawan from './Komponen/KartuKaryawan'; // Baris ini sangat
penting!

function App() {
```

```
  return (
    <div className="container mt-5">
      {/* BAGIAN MONITORING MESIN (Langkah 4 + Latihan 1 & 2) */}
      <h1 className="text-center mb-4">Dashboard Monitoring Produksi</h1>
      <div className="row mb-5">
        <div className="col-md-4">
          <KartuMesin nama="Mesin Injeksi-01" status="Running" produksi={150}
/>
        </div>
        <div className="col-md-4">
          <KartuMesin nama="Mesin CNC-02" status="Maintenance" produksi={45}
/>
        </div>
        <div className="col-md-4">
          <KartuMesin nama="Press-Hydraulic-05" status="Stop" produksi={85}
/>
        </div>
      </div>

      <hr />

      {/* TUGAS PROYEK MINI: BAGIAN KARYAWAN */}
      <h2 className="text-center my-4">Daftar Karyawan Aktif</h2>
      <div className="row">
        <div className="col-md-4">
          <KartuKaryawan nama="Budi Santoso" jabatan="Manager"
bagian="Produksi" />
        </div>
        <div className="col-md-4">
          <KartuKaryawan nama="Siti Aminah" jabatan="Operator"
bagian="Assembly" />
        </div>
        <div className="col-md-4">
          <KartuKaryawan nama="Agus Pratama" jabatan="QC" bagian="Quality
Control" />
        </div>
      </div>
    </div>
  );
}

export default App;
```



Gambar F.3 – Menampilkan hasil akhir aplikasi berupa kartu mesin/karyawan yang dibuat menggunakan komponen React dan props.

G. PEMBAHASAN

G.1 Analisis Kesesuaian Hasil dengan Teori

Hasil praktikum yang diperoleh sudah sesuai dengan teori yang dipelajari pada modul React. Dalam teori dijelaskan bahwa React menggunakan konsep component-based dan props untuk mengirim data, dan hal tersebut sudah berhasil diterapkan dalam pembuatan komponen KartuKaryawan.

Dari hasil output yang ditampilkan di browser, terlihat bahwa setiap komponen mampu menampilkan data yang berbeda meskipun menggunakan struktur yang sama. Hal ini menunjukkan bahwa penggunaan props berjalan dengan baik.

Selain itu, penggunaan JSX juga sudah sesuai dengan konsep yang dipelajari, yaitu penulisan tampilan UI menggunakan sintaks yang menyerupai HTML di dalam JavaScript. Dengan demikian, dapat disimpulkan bahwa hasil praktikum telah sesuai dengan teori baik dari segi konsep maupun implementasi.

G.2 Kendala yang Ditemukan dan Solusinya

Kendala / Error yang Ditemukan	Solusi yang Diterapkan
Terjadi error saat menjalankan project karena versi Node.js tidak sesuai	Menggunakan versi Vite yang kompatibel atau menyesuaikan versi Node.js
Komponen tidak muncul di browser	Memastikan file sudah di-import dengan path yang benar dan menggunakan export default
Tampilan tidak rapi	Menambahkan link Bootstrap pada file index.html

G.3 Kaitan dengan Penerapan di Industri

Keterampilan yang dipelajari pada praktikum ini sangat relevan dengan penerapan di dunia industri, khususnya dalam pengembangan sistem informasi berbasis web.

Sebagai contoh, dalam sebuah perusahaan manufaktur, React dapat digunakan untuk membuat dashboard monitoring karyawan atau produksi. Setiap data karyawan dapat ditampilkan menggunakan komponen seperti KartuKaryawan, sehingga informasi seperti jabatan dan bagian kerja dapat ditampilkan secara terstruktur.

Selain itu, konsep reusable component sangat membantu dalam pengembangan sistem skala besar, karena developer tidak perlu membuat ulang tampilan yang sama. Hal ini dapat meningkatkan efisiensi kerja dan mempermudah maintenance sistem.

H. KESIMPULAN

Tuliskan 3–5 poin kesimpulan yang merangkum hal-hal yang dipelajari dan dicapai pada praktikum modul ini. Kesimpulan harus spesifik, bukan bersifat umum.

1	React memungkinkan pembuatan tampilan antarmuka yang lebih terstruktur dengan konsep component-based.
2	Props digunakan untuk mengirim data antar komponen sehingga tampilan menjadi dinamis.
3	JSX mempermudah penulisan UI karena sintaksnya mirip dengan HTML.
4	Komponen yang dibuat dapat digunakan berulang kali sehingga lebih efisien.
5	Kendala teknis seperti versi Node.js dan import file dapat diatasi dengan penyesuaian konfigurasi.

I. DAFTAR PUSTAKA

No.	Referensi (Format APA 7th Edition)
[1]	Burhandenny, A. E. (2026). Manual Praktikum Aplikasi Web dan Mobile Semester Genap 2025/2026. Program Studi Teknik Industri, Universitas Negeri Yogyakarta.
[2]	Meta Platforms, Inc. (2024). *React Documentation*. https://react.dev
[3]	Vite. (2024). *Vite Guide*. https://vitejs.dev/guide/
[4]	Mozilla Developer Network. (2024). *JavaScript Guide*. https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide

J. LEMBAR PENILAIAN DAN PENGESAHAN

RUBRIK PENILAIAN LAPORAN					
Aspek Penilaian		Bobot			
No.	Aspek Penilaian	Bobot (%)	Nilai (0-100)	Skor Akhir	Catatan Penilai
1	Kelengkapan Isi (Semua bagian A–I terisi lengkap)	20%			
2	Kualitas Dasar Teori (Ditulis dengan bahasa sendiri, relevan, minimal 3 paragraf)	15%			
3	Dokumentasi Langkah Kerja (Kode & Screenshot sesuai, jelas, dan terbaca)	25%			
4	Tugas/Latihan Mandiri (Semua latihan & proyek mini dikerjakan dan berfungsi)	20%			
5	Pembahasan & Analisis (Kritis, mendalam, dan relevan dengan industri)	15%			
6	Kesimpulan & Tata Bahasa (Spesifik, rapi, mengikuti format yang ditentukan)	5%			
TOTAL NILAI AKHIR		100%			

Mahasiswa	Diperiksa oleh Asisten / Dosen
Syalfa Oktifa Nanda NIM: 23051430023	_____ Dr. Eng. Ir. Aji Ery Burhandenny, S.T., M.AIT. Tanggal: _____

PANDUAN DAN KETENTUAN PENGISIAN LAPORAN

Baca dengan seksama sebelum mengisi template laporan ini

1. Aturan Umum

a.	Laporan dikerjakan secara INDIVIDUAL, kecuali ada instruksi khusus dari dosen.
b.	Isi laporan harus merupakan hasil pekerjaan sendiri. Plagiarisme (menyalin laporan teman secara identik) akan mendapatkan nilai NOL (0) untuk kedua pihak.
c.	Bagian Dasar Teori HARUS ditulis dengan kalimat dan bahasa sendiri, bukan disalin mentah dari modul atau internet.
d.	Screenshot WAJIB memperlihatkan nama/NIM mahasiswa pada salah satu bagian tampilan (bisa di judul halaman web atau bagian lain) sebagai bukti keaslian pekerjaan.
e.	Semua teks dalam kotak yang berwarna abu-abu/biru muda dan bersifat miring adalah instruksi yang HARUS diganti dengan isi yang sesungguhnya.

2. Format File dan Ketentuan Pengumpulan

Hal	Ketentuan
Format File	Microsoft Word (.docx). Konversi ke PDF jika diminta dosen.
Nama File	Laporan_Modul[No]_[NIM]_[NamaLengkap].docx Contoh: Laporan_Modul02_12320001_AhmadDhani.docx
Ukuran Kertas	A4 (210 × 297 mm)
Margin	Atas: 3 cm | Bawah: 3 cm | Kiri: 4 cm | Kanan: 3 cm
Font Teks Isi	Arial, ukuran 11pt, spasi 1.15
Font Kode Program	Courier New, ukuran 10pt, latar gelap (sudah tersedia di template)
Batas Pengumpulan	7 hari kalender setelah tanggal pelaksanaan praktikum
Media Pengumpulan	Github repository mahasiswa + konfirmasi email ke dosen
Subjek Email	[WebLab-No.Modul] NIM - Nama Lengkap Contoh: [WebLab-02] 12320001 - Ahmad Dhani

3. Kompilasi Laporan Akhir Semester

PENTING – LAPORAN AKHIR SEMESTER

Pada akhir semester, seluruh laporan modul (Modul 1–12, kecuali Modul 13 UTS dan Modul 16 UAS) akan dikompilasi menjadi SATU BERKAS LAPORAN AKHIR. Ketentuan kompilasi:

a.	Laporan setiap modul dicetak/disimpan sebagai satu dokumen terpisah sesuai template ini.
b.	Kompilasi disatukan dengan urutan: Halaman Cover Akhir → Daftar Isi → Laporan Modul 1 → Laporan Modul 2 → … → Laporan Modul 12 → Laporan Modul 14 & 15.

c.	Laporan UTS (Modul 13) dan UAS (Modul 16) dikumpulkan secara TERPISAH dengan ketentuan yang diumumkan saat ujian.
d.	Setiap laporan modul di dalam kompilasi diawali dengan halaman Cover (Bagian A di template ini) sebagai pemisah antar modul.
e.	Nama file laporan akhir: LaporanAkhir_[NIM]_[NamaLengkap].docx
f.	Laporan akhir dikumpulkan paling lambat H-3 sebelum UAS.

4. Ketentuan Pengumpulan via GitHub

```
PraktikumWebMob_TekInd_2026/
├── README.md
├── Laporan/
│   ├── Laporan_Modul01_[NIM].docx
│   ├── Laporan_Modul02_[NIM].docx
│   └── ... (dst hingga Modul 15)
├── Pertemuan01_[NIM]_[Nama]/
│   ├── index.html
│   └── style.css
├── Pertemuan02_[NIM]_[Nama]/
└── ... (dst)
```

Pastikan setiap Commit memiliki pesan yang deskriptif. Contoh: "Menambahkan Laporan Modul 02 – CSS Dasar". Lakukan Push ke GitHub setiap akhir praktikum.

5. Konsekuensi Keterlambatan Pengumpulan

Keterlambatan	Konsekuensi
1–2 hari	Pengurangan nilai 10% dari total skor laporan
3–4 hari	Pengurangan nilai 25% dari total skor laporan
5–7 hari	Pengurangan nilai 50% dari total skor laporan
> 7 hari	Laporan TIDAK DITERIMA (nilai 0) – wajib temui dosen untuk klarifikasi
Tidak dikumpulkan	**Nilai laporan = 0, mahasiswa tidak diperkenankan mengikuti UAS**

— Selamat mengerjakan —